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Democratizing financial knowledge with ChatGPT by OpenAI: Unleashing the Power of Technology

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Abstract

This paper discusses the potential of using Explainable AI (XAI) and language models, such as ChatGPT, in revolutionizing the communication of financial knowledge to non-financial professionals. We tested the capabilities of ChatGPT in interpreting XAI models and explaining complex financial concepts in a non-technical manner. The results show that ChatGPT has great potential as a tool for making complex financial concepts accessible to a wide range of target audiences. To the best of the authors' knowledge, we are the first to present an approach to combine the method of Explainable AI (XAI) using SHAP with a conversational AI model, ChatGPT, to explain the predictions made by a nonlinear black box machine learning model in Finance. This paper also identifies the limitations of ChatGPT for explaining model predictions, but it is expected that the improvement of the prompt engineering aspect of the model will overcome these limitations. Ultimately, the use of ChatGPT has the potential to level the playing field and empower all individuals, regardless of their financial background, to make informed investment decisions.

Keywords: *finance and technology, investment, inequality, trading, ChatGPT, OpenAI, explainable AI, SHAP, XAI, black box model, machine learning, trading strategy, hedge fund, quantitative trading, factor loadings, style factor*

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Disclaimer

"The authors wrote this paper in part with GPT-3.5, OpenAI's large-scale language-generation model. Upon generating draft language, the author reviewed, edited, and revised the language to their own liking and takes ultimate responsibility for the content of this publication. Please advised that the contents of this paper should not be construed as investment advice."

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Introduction

The use of technology in the investment industry has the potential to democratize access to investment, particularly for individuals who may have been previously excluded due to financial constraints or lack of knowledge. One way this can be achieved is through the use of robo-advisors, which use algorithms to provide investment advice and manage portfolios at a lower cost than traditional human financial advisors.

The recent unveiling of ChatGPT by OpenAI represents a seminal development in the realm of language model AI, possessing the potential to revolutionize the financial industry. This technology can be employed to furnish investment advice and information to individuals through a conversational format, thereby making investment information more accessible and comprehensible to a broader range of individuals, particularly non-accredited investors and non-financial professionals who have historically been excluded from access to alternative investment products, such as hedge funds, as they are lacking the resources and expertise to "fend for themselves".

By offering personalized investment advice through a conversational interface, ChatGPT by OpenAI has the potential to provide a cost-effective method to address the issue of information asymmetric in financial investment and help to level the playing field for individual investors.

This study attempts to evaluate the extent to which ChatGPT can convert financial models into a user-friendly conversational format that can be easily comprehended by individuals with varying levels of financial literacy. In particular, this paper attempts to examine the capabilities of explainable AI and ChatGPT in explaining complex financial results. Financial modeling like the machine learning-based stock return model that incorporates multiple factors often involves intricate mathematical and statistical techniques, making it challenging for non-financial professionals to interpret the results. However, the use of explainable AI and ChatGPT has the potential to overcome these difficulties by providing clear and concise explanations of the results generated by an AI-based model. In Part III of the paper, we will discuss the implications and findings of our test cases.

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I. Technology and Access to Investment

According to Lusardi and Mitchell (2014), numerous studies have demonstrated the prevalence of low financial literacy among individuals and the resulting challenges they face in navigating complex financial and credit markets that require informed decision-making.[5] The lack of financial knowledge among unsophisticated consumers often leaves them vulnerable to financial scams and is deemed necessary to implement protective legislation.[6] This issue has gained increased attention in the aftermath of the financial crisis and efforts have been made to protect unsophisticated investors.[7] As an example, Lusardi and Mitchell highlight the Dodd-Frank Act of 2010 and the establishment of the U.S. Consumer Financial Protection Bureau, which aimed to provide better consumer protection and establish uniform standards for financial products.[8]

When Lusardi and Mitchell (2014) wrote the paper, it is recognised that it is difficult to cost-effective ways to build financial knowledge.[9] The alternative solutions they proposed were to help people simplify their investment decision for the non-financial professionals or outsource the works to financial advice.

However, the advance of technological applications in finance will offer a third solution.

Technology has been recognized as a key enabler of financial literacy and financial inclusion for the general public.[10] David Mhlanga argued that AI exerts a significant impact on digital financial inclusion in several domains, such as the detection and evaluation of risk, the alleviation of information asymmetry, the availability of customer support and helpdesk through chatbots, and the detection and deterrence of fraud and cybersecurity incidents.[11] The integration

[5] Lusardi, Annamaria, and Olivia S. Mitchell. 2014. "The Economic Importance of Financial Literacy: Theory and Evidence." Journal of Economic Literature 52 (1): 34. http://dx.doi.org/10.1257/jel.52.1.5
[6] *ibid*.
[7] *ibid*.
[8] *ibid*.
[9] *ibid*., 38.
[10] OECD. 2021. Digital delivery of financial education: design and practice. www.oecd.org/financial/education/digital-delivery-of-financial-education-design-and-practice.htm
[11] Mhlanga, David. 2020. "Industry 4.0 in finance: the impact of artificial intelligence (AI) on digital financial inclusion." International Journal of Financial Studies 8 (3): 45. https://doi.org/10.3390/ijfs8030045

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of technology such as AI, Machine learning (ML) and natural language processing (NLP) in the financial services sector has provided individuals with unprecedented access to information, enabling them to make informed decisions regarding their investments.[12]

For example, AI has enabled banks to incorporate customer support and help desks with Electronic Virtual Assistants (EVA), thereby leading to a more efficient and cost-effective delivery of customer support.[13] Furthermore, through the application of AI, financial institutions are able to offer personalized banking services, where chatbots and AI assistants use AI algorithms to generate individualized financial advice and utilize natural language processing for immediate self-service customer support.[14]

The reduction in the cost of acquiring financial education or information leads to enhanced access to investment opportunities for individuals who are not financial professionals or accredited investors. Under the current regulatory framework, only the only accredited investors[15], defined as those persons "whose financial sophistication and ability to sustain the risk of loss of investment or ability to fend for themselves render the protections of the Securities Act's registration process unnecessary", were eligible to invest in certain financial products, including hedge funds. [16] This posed a significant barrier for non-accredited investors, who were often excluded from higher-yielding investments. Thus, the present research attempts to investigate whether ChatGPT is capable of providing non-accredited investors at different education levels with financial knowledge derived from both simple and complex models, thereby mitigating information asymmetry in the investment process.

[12] Aroomoogan, Kumesh. 2020. "How Artificial Intelligence Can Enable Ethically Driven Investments." Forbes, September 23, 2020, https://www.forbes.com/sites/forbesfinancecouncil/2020/09/23/how-artificial-intelligence-can-enable-ethically-driven-investments/?sh=168099eb6fc1.
[13] *ibid*.
[14] *ibid*.
[15] Securities and Exchange Commission. 1987. "Regulation D Revisions; Exemption for Certain Employee Benefit Plans." Release No. 33-6683. January 16. [52 FR 3015].
[16] Regulation D

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II. Test cases

1. Simple Model

As a first test case, we demonstrate the use of ChatGPT in explaining definition of fundamental financial metrics. The abundance of data and academic literature that ChatGPT has been trained on, allows for the potential for it to effectively explain the intricacies of various factor definitions. This can be particularly useful in finance, as understanding the fundamental financial metrics is a key step for making informed investment decisions. Additionally, we aim to investigate the impact of tone and target audience on the language chosen by a language model such as ChatGPT. Building on the concept of the "mom's test", which posits that the ability to explain complex financial concepts in a manner that is easily understandable by a layperson is an important aspect of effective communication, we aim to explore whether varying the tone and target audience can lead to alternative word choices that may be more suitable for different audiences.[17] Specifically, we will examine the use of ChatGPT to generate explanations of financial concepts in different tones, with the goal of determining whether this approach can improve the effectiveness of communication for different audiences.

The tests cover a range of finance concepts, starting with basic ones (beta, alpha) and progressing to more advanced ones (bid-ask spread, idiosyncratic return volatility, illiquidity, earnings-to-price). The ability of GPT to generate relevant explanations with minimal prompts (i.e., just the factor name, authors, publication year, and journal name) is also examined.

a) Alpha

Prompt: Explain the meaning of alpha in finance.

ChatGPT[18]: In finance, alpha is a measure of a portfolio's or investment's performance in relation to a benchmark index. It represents the excess return of the investment

[17] Fitzpatrick, Rob. 2013. The Mom Test: How to talk to customers & learn if your business is a good idea when everyone is lying to you. Robfitz Ltd.
[18] OpenAI. 2022. "ChatGPT: Optimizing Language Models for Dialogue." Accessed February 01, 2023. https://openai.com/blog/chatgpt/.

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relative to the return of the benchmark index. A positive alpha indicates that the investment has performed better than the benchmark, while a negative alpha indicates that the investment has underperformed the benchmark. Alpha is often used as a measure of a portfolio manager's skill and is an important metric in evaluating investment performance.

Prompt: Explain to non-finance professionals what alpha is in finance.

ChatGPT[19]: In finance, alpha is a measure of a portfolio or investment manager's performance relative to a benchmark index, such as the S&P 500. It represents the excess return of an investment compared to what would have been expected given the level of risk taken. A positive alpha indicates that the investment has performed better than expected,
while a negative alpha indicates that it has performed worse. In other words, it's a measure of the returns generated by a portfolio or investment manager, that is not explained by the market returns. It's a measure of the manager's skill in selecting investments.

Prompt: Explain to a high school student what alpha is in finance.

ChatGPT[20]: In finance, alpha is a measure of the performance of an investment relative to a benchmark, like the stock market. It represents the return on an investment that is not due to the overall market movement. In other words, it is the extra return that an investment earns over and above the return of the market. For example, if a stock has an alpha of 2%, that means it has performed 2% better than the market. A positive alpha means the investment has performed better than the benchmark, while a negative alpha means it has performed worse. High school students can think of alpha as a grade or score given to an investment, with a higher score being better.

[19] *Ibid.*
[20] *ibid.*

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b) Beta

Prompt: Explain the meaning of beta in finance.

ChatGPT[21]: Beta in finance is a measure of a stock's or portfolio's volatility in relation to the overall market. It is used to determine the level of risk associated with an investment. A beta of 1 means that the stock or portfolio's price will move with the market. A beta less than 1 means it is less volatile than the market, while a beta greater than 1 means it is more volatile. It is an important metric used in the Capital Asset Pricing Model (CAPM) to calculate the expected return of an investment. Beta can also be used to compare the volatility of different investments, with higher beta investments typically considered riskier than lower beta investments.

Prompt: Explain to non-finance professionals what beta is in finance.

ChatGPT[22]: Beta is a measure of a stock's volatility in relation to the overall market. A beta of 1 means that the stock's price will move with the market, while a beta less than 1 means it is less volatile than the market, and a beta greater than 1 means it is more volatile.

For example, if a stock has a beta of 1.5, it is 50% more volatile than the market. If the market goes up by 10%, the stock is expected to go up by 15%. Similarly, if the market goes down by 10%, the stock is expected to go down by 15%.

In general, a stock with a higher beta is considered to be riskier than a stock with a lower beta, because it is more affected by market movements. However, high beta stocks also have the potential for higher returns. Beta is an important factor to consider when constructing a diversified portfolio and assessing risk.

Prompt: Explain to a high school student what beta is in finance.

[21] *ibid.*
[22] *ibid.*

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ChatGPT[23]: Beta in finance is a measure of a stock's volatility in relation to the overall market. It is used to help assess the risk of a stock, with a beta of 1 indicating that the stock's price will move with the market, a beta less than 1 indicating that it is less volatile than the market, and a beta greater than 1 indicating that it is more volatile. In simpler terms, beta is a way to measure how much a stock's price goes up and down compared to the market as a whole. For high school students, it can be helpful to think of beta as a tool that helps investors understand the level of risk associated with a particular stock. [18]

c) Earnings to Price

Prompt: Factor: ep, Earnings to price, Basu, 1977, JF less than 100 words

ChatGPT[24]: Earnings to price (EP) is a financial ratio that compares a company's earnings per share (EPS) to its stock price. The ratio is calculated by dividing the EPS by the current stock price. EP is often used to evaluate a company's financial performance and to make investment decisions. A high EP ratio indicates that a company's earnings are high relative to its stock price, which may suggest that the stock is undervalued. A low EP ratio, on the other hand, may suggest that the stock is overvalued.

Prompt: Explain to non-finance professionals.

ChatGPT[25]: Earnings to Price, or EP, is a financial ratio that compares a company's earnings per share (EPS) to its stock price. It is used to evaluate a company's profitability and assess whether its stock is overvalued or undervalued. A high EP ratio indicates that the company is profitable and that its stock may be undervalued, while a low EP ratio suggests that the company is less profitable and that its stock may be overvalued. The EP ratio was first introduced by Sanjoy Basu in a 1977 paper published in the Journal of Financial Economics. It is often used in conjunction with other financial ratios, such as price-to-earnings (P/E) and price-

[23] *ibid.*
[24] *ibid.*
[25] *ibid.*

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to-sales (P/S), to gain a more comprehensive understanding of a company's financial performance.

Prompt: Explain to my mom.

ChatGPT[26]: EP is a way of measuring how much a company is making compared to how much its stock is worth. It's like looking at how much money a store makes compared to how much people are willing to pay for the things they sell. A company with a high EP is making a lot of money compared to how much its stock is worth. A company with a low EP is not making as much money compared to how much its stock is worth. This is something that investors can use to help them decide if they want to buy stock in a company or not.

d) Illiquidity

Prompt: ill, Illiquidity Amihud, 2002, JFM[27]

ChatGPT[28]: The "ill" factor, also known as the Illiquidity factor, is a measure of how difficult it is to buy or sell a stock without affecting its price. It was first introduced by Amihud in 2002 in the Journal of Financial Markets. It is calculated by taking the ratio of the stock's absolute return over the stock's dollar volume. The higher the illiquidity, the more difficult it is to trade the stock, and the higher the risk associated with investing in it.

Prompt 1: Explain to non-finance professionals.

ChatGPT[29]: illiquidity, refers to the difficulty of buying or selling an asset, such as a stock or bond, due to a lack of market participants or a lack of interest from buyers or sellers. Illiquidity can make it difficult for investors to enter or exit a position in a security and can also make it difficult for investors to value their

[26] *ibid.*
[27] Amihud, Yakov. 2002. "Illiquidity and Stock Returns: Cross-Section and Time-Series Effects." Journal of Financial Markets 5, no. 1: 31-56.
[28] OpenAI. 2022.
[29] *ibid.*

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investments. Amihud introduced the concept of Illiquidity in 2002 in the Journal of Financial Markets.

Prompt 2: Explain to my mom.

ChatGPT[30]: "Illiquidity" is a measure of how easy or difficult it is to buy or sell a stock. If a stock is considered illiquid, it means that there aren't a lot of buyers or sellers, so it can be harder to find someone to trade with. This can make the price of the stock more volatile, and also make it more expensive to buy or sell. Amihud 2002 published this concept in the Journal of Financial Markets.

2. Complicated Model

Factor investing uses specific characteristics, or factors, to explain returns and risks of a group of securities. Fama and French proposed an extension of the Capital Asset Pricing Model (CAPM) by incorporating two additional factors, size, and value, to capture the characteristics of common stock returns. [31] In the hedge fund industry, practitioners identify and isolate specific factors that have historically exhibited a persistent risk premium as part of their systematic approach to capturing alpha, which is often considered an alternative to traditional stock picking.

In recent years, the use of many factors, known as the "factor zoo," has been a source of concern in the investment community. This is due to the lack of interpretability and transparency associated with this approach. Research by Feng, Giglio, and Xiu has shown that many factors have little impact on returns and are highly correlated with one another, making it difficult to identify their individual contributions to performance.[32] The importance of addressing the factor zoo problem and developing human interpretable investment models is critical to the success of hedge funds and the overall integrity of the financial industry.

[30] *ibid.*
[31] Fama, Eugene F., and Kenneth R. French. 1993. "Common risk factors in the returns on stocks and bonds." Journal of Financial Economics 33, no. 1: 3-56.
[32] Feng, Guanhao, Stefano Giglio, and Dacheng Xiu. 2020. "Taming the Factor Zoo: A Test of New Factors." The Journal of Finance 75, no. 3: 1327-1370.

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Explainable AI (XAI) is an area of research that focuses on creating AI systems that can be understood and interpreted by humans. The goal of explainable AI is to increase transparency and accountability of AI systems, which is particularly important in high-stakes domains such as finance and healthcare. In the context of investment, explainable AI can play a crucial role in solving the factor zoo problem, which refers to the difficulty of determining the underlying factors that drive the returns of hedge funds and other complex investment strategies.

Lundberg, S. M., & Lee, S. I. introduced the innovative approach of using SHapley Additive exPlanations (SHAP) values in the field of explainable AI. [33] [34] SHAP values offer a comprehensive and uniform method of attributing the predictions of any machine learning model to its input features. Rooted in the principles of Shapley values from cooperative game theory, SHAP values ensure a just distribution of value among a group of players. In the context of investment, SHAP values can be used to attribute the returns of a hedge fund or other investment strategy to its underlying factors.

Combining XAI methods such as SHAP with natural language processing tools like ChatGPT can add significant value in terms of explaining factor attributions. One of the main advantages of this approach is that it allows investors to understand the rationale behind the factor attributions, rather than just being presented with a blackbox. This is particularly important in the context of factor investing and hedge fund replication, as investors often struggle to understand the underlying drivers of returns. Another key advantage of this approach is that it can help to mitigate the problem of the "factor zoo" by providing investors with a clear, human-understandable explanation regarding which factors are driving returns.

Additionally, by combining XAI methods and natural language processing tools, investors can gain a more holistic view of their portfolio or stock level factor attribution. This can help to identify any potential risks or opportunities that might be obscured by a more traditional, blackbox approach.

[33] Lundberg, Scott M., Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. 2020. "From local explanations to global understanding with explainable AI for trees." Nature Machine Intelligence 2, no. 1: 56-67.
[34] Lundberg, Scott M., and Su-In Lee. 2017. "A unified approach to interpreting model predictions." In Advances in Neural Information Processing Systems 30.

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Furthermore, this approach has the potential to improve the transparency of the investment process, which can be beneficial for both investors and asset managers. By providing clear, human-understandable explanations of the factor attributions, it becomes much easier to detect any potential issues or errors that might be present in the investment process. This can help to increase trust in the investment process and can lead to more effective decision-making.

The purpose of this study is to evaluate the feasibility of providing more accessible and interpretable explanations of the predictions made by a complex AI model. We utilized an AI model[35] that was designed to forecast the top 10% of SP500 stock performance in the upcoming 3 months implemented by MechaniX Limited using data from Refinitiv up to the January 24th, 2023, market close. The model identified "Norwegian Cruise Line Holdings Ltd" (NCLH.N) as a potential top 10% performer among SP500 constituents. It is important to note that this work focuses on the integration of ChatGPT and XAI, and as such, the exact method of stock prediction and its accuracy are not the primary focus of this research.

To gain insight into the underlying factors that contributed to this prediction, we employed the SHAP methodology on the model, which generated a waterfall plot. The X-axis of the plot displays the log-odds unit or the SHAP value, while the Y-axis illustrates the short name of the style factors utilized as features in the prediction, along with their corresponding values.

[35] Gu, Shihao, Bryan Kelly, and Dacheng Xiu. 2020. "Empirical Asset Pricing via Machine Learning." The Review of Financial Studies 33, no. 5: 2223-2273.

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Figure 1: Waterfall chart for black box AI model[36]

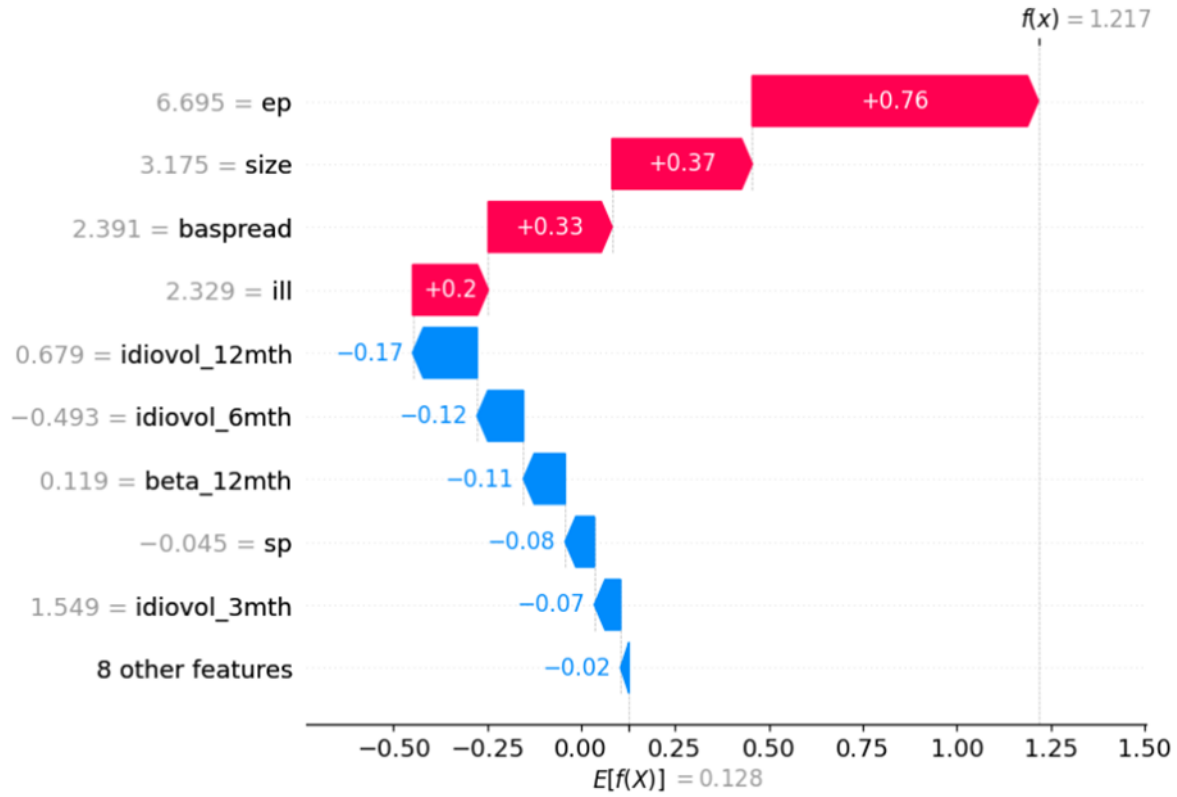


Figure 1. A waterfall plot generated by SHAP package by MechaniX Limited to explain the features importance of AI model for Norwegian Cruise Line Holdings.

The resulting waterfall plot can be difficult for individuals without a background in machine learning, explainable AI, or quantitative analysis to fully comprehend (see Figure 1).

In order to improve the interpretability of the results for a wider audience, we employed a technique that utilized the output from our SHAP explanation was transformed into ChatGPT prompts in an automated fashion. The process can be fully automated as the information from the waterfall plot can easily be converted into a "prompt" through a prompt template. To use the template, we fill in the information to all the angle bracketed cells according to the description. In case of a more complicated template, we can leverage other more advanced techniques like meta-language e.g., Backus-Naur Form[37] to define it.

[36] Generated by MechaniX Limited, http://mechanix.xyz
[37] Knuth, Donald E. 1964. "Backus normal form vs. backus naur form." Communications of the ACM 7, no. 12: 735-736.

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Prompt Template:

The following report aims to provide an explanation of why the model has predicted the stock (<***Company Full Name***>) <***Company Ticker/RIC Code***> to have the highest chance to out-platform 90% of stocks among the SP500 in the next 3 months based on the factor exposure and Shapley value. The factors exposures and their corresponding Shapley value are as follows:

1. <***1st important factor***> (<***short name for the 1st factor***>) – factors exposures = <***factor exposure for the 1st factor***>, Shapley value = <***Shapley value for the 1st factor***>
2. <***2nd important factor***> (<***short name for the 2nd factor***>) – factors exposures = <***factor exposure for the 2nd factor***>, Shapley value = <***Shapley value for the 2nd factor***>
3. <***3rd important factor***> (<***short name for the 3rd factor***>) – factors exposures = <***factor exposure for the 3rd factor***>, Shapley value = <***Shapley value for the 3rd factor***>
4. <***4th important factor***> (<***short name for the 4th factor***>) – factors exposures = <***factor exposure for the 4th factor***>, Shapley value = <***Shapley value for the 4th factor***>
5. <***5th important factor***> (<***short name for the 5th factor***>) – factors exposures = <***factor exposure for the 5th factor***>, Shapley value = <***Shapley value for the 5th factor***>
6. <***6th important factor***> (<***short name for the 6th factor***>) – factors exposures = <***factor exposure for the 6th factor***>, Shapley value = <***Shapley value for the 6th factor***>
7. <***7th important factor***> (<***short name for the 7th factor***>) – factors exposures = <***factor exposure for the 7th factor***>, Shapley value = <***Shapley value for the 7th factor***>
8. <***8th important factor***> (<***short name for the 8th factor***>) – factors exposures = <***factor exposure for the 8th factor***>, Shapley value = <***Shapley value for the 8th factor***>
9. <***9th important factor***> (<***short name for the 9th factor***>) – factors exposures = <***factor exposure for the 9th factor***>, Shapley value = <***Shapley value for the 9th factor***>

The factor of <***1st important factor***> (<***short name for the 1st factor***>) has the highest Shapley value of <***Shapley value for the 1st factor***>, indicating that it has the strongest impact on the prediction outcome. The factor of <***2nd important factor***> (<***short name for the 2nd factor***>) has the second highest Shapley value of <***Shapley value for the 2nd factor***>, followed by <***3rd important factor***> (<***short name for the 3rd factor***>) with Shapley value of <***Shapley value for the 3rd factor***>.
Please explain it like an article in Barron's magazine but do not use the term Shapley value as this term is too technical. Try to explain it purely using financial concepts and terms.

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a) Explain Black Box AI Model prediction like Barron's.

Prompt 1 (Explain like Barron's):

Prompt: The following report aims to provide an explanation of why the model has predicted the stock

(Norwegian Cruise Line Holdings Ltd) NCLH.N to have the highest chance to out-platform 90% of stocks among the SP500 in the next 3 months based on the factor exposure and Shapley value. The factors exposures and their corresponding Shapley value are as follows:

1. Earnings to price (ep) – factors exposures = 6.695, Shapley value = 0.76
2. Size (size) – factors exposures = 3.175, Shapley value = 0.37
3. Bid-ask spread (baspread) – factors exposures = 2.391, Shapley value = 0.33
4. Illiquidity (ill) – factors exposures = 2.329, Shapley value = 0.2
5. Idiosyncratic return volatility (idiovol_12mth) – factors exposures = 0.679, Shapley value = -0.17
6. Idiosyncratic return volatility (idiovol_6mth) – factors exposures = -0.493, Shapley value = -0.12
7. Beta (beta_12mth) – factors exposures = -0.045, Shapley value = -0.08
8. Market return (sp) – factors exposures = -0.045, Shapley value = -0.08
9. Idiosyncratic return volatility (idiovol_3mth) -- factors exposures = 1.549, Shapley value = -0.07

The factor of Earnings to Price (ep) has the highest Shapley value of 0.76, indicating that it has the strongest impact on the prediction outcome. The factor of Size (size) has the second highest

Shapley value of 0.37, followed by Bid-ask spread (baspread) with Shapley value of 0.33.

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Please explain it like an article in Barron's magazine but do not use the term Shapley value as this term is too technical. Try to explain it purely using financial concepts and terms.

ChatGPT[38]: Norwegian Cruise Line Holdings Ltd (NCLH.N) is predicted to be among the top performers in the S&P 500 index in the next 3 months, according to a recent analysis by our model. The model uses a combination of financial metrics and factors to make its predictions.

One of the key factors that sets NCLH.N apart is its Earnings to Price (ep) ratio. This ratio compares the company's earnings to the price of its stock, providing insight into the company's profitability and valuation. The model's analysis shows that NCLH.N has a particularly strong ep ratio, which is likely contributing to its predicted strong performance.

Another important factor that the model is taking into account is the company's size. The model looks at the company's market capitalization, which reflects its overall size and potential for growth. NCLH.N's market capitalization is considered to be a positive factor in the model's prediction.

Finally, the model is also considering the liquidity of NCLH.N's shares. The Bid-ask spread and Illiquidity measures the ease with which shares can be bought or sold in the market. These factors are considered as positive factors in the model's prediction.

It is worth noting that the model is also considering the negative factors such as Idiosyncratic return volatility, Beta and Market return which have negative effect on the prediction. This suggests that the model is considering the volatility of the stock's returns and the level of risk associated with investing in the company.

[38] *ibid.*

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Overall, the model's analysis suggests that NCLH.N's strong earnings to price ratio, size, and liquidity are contributing to its expected gains in the next 3 months. However, as with any investment, it is important to conduct thorough research and analysis before making any decisions. Additionally, it is important to take into account the current market conditions, economic outlook, and other external factors that could impact the performance of the stock.

Prompt 2 (Explain like Financial Times):

Prompt: The following report aims to provide an explanation of why the model has predicted the stock

(Norwegian Cruise Line Holdings Ltd) NCLH.N to have the highest chance to out- platform 90% of stocks among the SP500 in the next 3 months based on the factor exposure and Shapley value. The factors exposures and their corresponding Shapley value are as follows:

1. Earnings to price (ep) – factors exposures = 6.695, Shapley value = 0.76
2. Size (size) – factors exposures = 3.175, Shapley value = 0.37
3. Bid-ask spread (baspread) – factors exposures = 2.391, Shapley value = 0.33
4. Illiquidity (ill) – factors exposures = 2.329, Shapley value = 0.2
5. Idiosyncratic return volatility (idiovol_12mth) – factors exposures = 0.679, Shapley value = -0.17
6. Idiosyncratic return volatility (idiovol_6mth) – factors exposures = -0.493, Shapley value = -0.12
7. Beta (beta_12mth) – factors exposures = -0.045, Shapley value = -0.08
8. Market return (sp) – factors exposures = -0.045, Shapley value = -0.08
9. Idiosyncratic return volatility (idiovol_3mth) -- factors exposures = 1.549, Shapley value = -0.07

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The factor of Earnings to Price (ep) has the highest Shapley value of 0.76, indicating that it has the strongest impact on the prediction outcome. The factor of Size (size) has the second highest

Shapley value of 0.37, followed by Bid-ask spread (baspread) with Shapley value of 0.33.

Please explain and expand it like an article from Financial Times but do not use the term Shapley value as this term is too technical. Try to explain it purely using financial concepts and terms.

ChatGPT[39]: The Norwegian Cruise Line Holdings Ltd (NCLH.N) has been identified as a top performer among the stocks in the S&P 500 index over the next three months, based on its exposure to various financial factors. This report seeks to explain why NCLH is expected to outperform 90% of the other stocks in the index.

According to the analysis, NCLH is particularly well-positioned in terms of its earnings to price ratio (ep). This ratio is a measure of how much profit a company generates for every unit of its stock price, and it is considered an important indicator of a stock's potential for growth. In the case of NCLH, its exposure to this factor is strong, with a score of 6.695, contributing significantly to its overall potential for growth.
Another important factor contributing to NCLH's expected performance is its size, measured by its market capitalization. Companies with a larger market cap are often more established and have more resources, making them more attractive to investors. NCLH has a size score of 3.175, placing it well ahead of many of its peers in the S&P 500.

Another factor that makes NCLH a strong investment opportunity is its bid-ask spread, or the difference between the highest price a buyer is willing to pay for a stock and the lowest price a seller is willing to accept. A smaller spread generally indicates more liquidity in the market, making it easier for investors to

[39] *ibid.*

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trade the stock. NCLH's bid-ask spread score of 2.391 puts it in a strong position in this regard.

Finally, it is worth mentioning that NCLH also has a relatively low score for idiosyncratic return volatility, which measures the unpredictability of a company's returns. A lower score in this category is generally seen as a positive, as it indicates that the stock is less likely to experience large, sudden price swings. This factor, along with others such as market return and beta, contribute to NCLH's overall strength as a potential investment opportunity.

In conclusion, NCLH's strong performance is based on its exposure to key financial factors such as earnings to price, size, bid-ask spread, and low idiosyncratic return volatility. These factors make it a compelling investment opportunity for those looking for growth over the next three months.

Prompt 3 (Explain like South China Morning Post)

Prompt: The following report aims to provide an explanation of why the model has predicted the stock

(Norwegian Cruise Line Holdings Ltd) NCLH.N to have the highest chance to out-platform 90% of stocks among the SP500 in the next 3 months based on the factor exposure and Shapley value. The factors exposures and their corresponding Shapley value are as follows:

1. Earnings to price (ep) – factors exposures = 6.695, Shapley value = 0.76
2. Size (size) – factors exposures = 3.175, Shapley value = 0.37
3. Bid-ask spread (baspread) – factors exposures = 2.391, Shapley value = 0.33
4. Illiquidity (ill) – factors exposures = 2.329, Shapley value = 0.2
5. Idiosyncratic return volatility (idiovol_12mth) – factors exposures = 0.679, Shapley value = -0.17

Electronic copy available at: https://ssrn.com/abstract=4346152

6. Idiosyncratic return volatility (idiovol_6mth) – factors exposures = -0.493, Shapley value = -0.12
7. Beta (beta_12mth) – factors exposures = -0.045, Shapley value = -0.08
8. Market return (sp) – factors exposures = -0.045, Shapley value = -0.08
9. Idiosyncratic return volatility (idiovol_3mth) -- factors exposures = 1.549, Shapley value = -0.07

The factor of Earnings to Price (ep) has the highest Shapley value of 0.76, indicating that it has the strongest impact on the prediction outcome. The factor of Size (size) has the second highest

Shapley value of 0.37, followed by Bid-ask spread (baspread) with Shapley value of 0.33.

Please explain it like an article from South China Morning Post but do not use the term Shapley value as this term is too technical. Try to explain it purely using financial concepts and terms.

ChatGPT[40]: The recent report suggests that Norwegian Cruise Line Holdings Ltd (NCLH.N) is predicted to have the highest change among the stocks in the S&P 500 in the next three months. The prediction is based on the company's exposure to various financial factors and their impact on the stock's performance.

The strongest factor affecting NCLH.N's performance is its earnings to price ratio, which measures the relationship between the company's earnings and its stock price. With an exposure score of 6.695, this factor is expected to have a significant impact on the company's stock performance.

Second in line is the company's size, with an exposure score of 3.175. This factor takes into account the company's market capitalization and the number of shares outstanding, providing an indication of its overall market presence.

[40] *ibid.*

Electronic copy available at: https://ssrn.com/abstract=4346152

The bid-ask spread, which measures the difference between the highest price a buyer is willing to pay for a stock and the lowest price a seller is willing to accept, is also expected to have a moderate impact on NCLH.N's performance, with an exposure score of 2.391.

Other factors affecting NCLH.N's performance include illiquidity, which measures how easily a stock can be bought or sold, and the company's idiosyncratic return volatility, which is a measure of the stock's risk not explained by the overall market. The report suggests that these factors will have a lesser impact on the company's stock performance.

In conclusion, the report predicts that NCLH.N will have a strong performance in the next three months, based on its exposure to various financial factors, with the earnings to price ratio being the strongest indicator of the company's expected performance.

Prompt 4 (Explain to my mother)

Prompt: The following report aims to provide an explanation of why the model has predicted the stock

(Norwegian Cruise Line Holdings Ltd) NCLH.N to have the highest chance to out-platform 90% of stocks among the SP500 in the next 3 months based on the factor exposure and Shapley value. The factors exposures and their corresponding Shapley value are as follows:

1. Earnings to price (ep) – factors exposures = 6.695, Shapley value = 0.76
2. Size (size) – factors exposures = 3.175, Shapley value = 0.37
3. Bid-ask spread (baspread) – factors exposures = 2.391, Shapley value = 0.33
4. Illiquidity (ill) – factors exposures = 2.329, Shapley value = 0.2
5. Idiosyncratic return volatility (idiovol_12mth) – factors exposures = 0.679, Shapley value = -0.17

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6. Idiosyncratic return volatility (idiovol_6mth) – factors exposures = -0.493, Shapley value = -0.12
7. Beta (beta_12mth) – factors exposures = -0.045, Shapley value = -0.08
8. Market return (sp) – factors exposures = -0.045, Shapley value = -0.08
9. Idiosyncratic return volatility (idiovol_3mth) -- factors exposures = 1.549, Shapley value = -0.07

The factor of Earnings to Price (ep) has the highest Shapley value of 0.76, indicating that it has the strongest impact on the prediction outcome. The factor of Size (size) has the second highest Shapley value of 0.37, followed by Bid-ask spread (baspread) with Shapley value of 0.33.

Please explain the above report to my mother without using any professional terms.

ChatGPT[41]: The report is talking about a stock called Norwegian Cruise Line Holdings Ltd and why a model thinks it might do really well in the next 3 months compared to other stocks. The model looks at different things about the stock, like how much money it makes and how big it is, to make this prediction. Out of all the things the model looks at, it thinks the most important one is how much money the stock makes compared to how much it costs. The next most important things are the size of the stock and how easy it is to buy and sell it. All of these things combined make the model think this stock has a good chance of doing well in the next few months.

III. Discussion

Regarding the simple model, the quality of the output generated by ChatGPT was acceptable. It was able to effectively transform financial knowledge for non-financial professionals, providing explanations of concepts such as alpha and beta to non-finance professionals and even high

[41] *ibid.*

Electronic copy available at: https://ssrn.com/abstract=4346152

school students. It is worth mentioning that when the target audience is not specified, ChatGPT tends to provide responses that are more technical in nature and may include terms and concepts that may not be easily understood by non-finance professionals. For instance, without specifying a target audience, ChatGPT may use terms such as "benchmark" or "alpha signals" that may not be familiar to individuals without a financial background. This highlights the importance of specifying the target audience when utilizing ChatGPT as a tool for disseminating financial information. By clearly defining the target audience, ChatGPT can adjust its language and terminology to be more accessible and easier to understand for individuals with limited financial knowledge, while still providing accurate and technical explanations for more advanced audiences.

The combination of XAI methods like SHAP and ChatGPT enabled us to generate a more accessible and easy-to-understand explanation, which can be presented in a format similar to that of a financial publication. By extracting the relevant information from the plot and inputting it into a "prompt" for ChatGPT, we were able to generate a more understandable explanation that could be presented in a manner similar to that of a financial news outlet such as Barron's, Financial Times, and South China Morning Post. We also tested the extreme case of generating an explanation to my “mother” without using any professional terms and the results are quite impressive. This approach demonstrates the potential for utilizing natural language generation techniques to improve the accessibility and comprehension of complex financial information for investors.

However, limitations in the utilization of ChatGPT for explaining model predictions were also noted. One such limitation is related to the bid-ask spread factor, where a higher score is typically indicative of a larger bid-ask spread, which is generally associated with a higher level of illiquidity risk that must be compensated by higher returns.[42] In some instances, ChatGPT has demonstrated an inadequate understanding of this relationship, and has even erroneously concluded that a lower bid-ask spread is a more favourable predictor of future stock returns.

[42] Amihud, Yakov, and Haim Mendelson. 1989. "The effects of beta, bid-ask spread, residual risk, and size on stock returns." The Journal of Finance 44 (2): 479-486.

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The ChatGPT model has shown great promise as a tool for financial engineering, but there remain limitations that must be addressed. Further research is needed to enhance the prompt engineering aspect of the model to overcome these obstacles. One promising avenue for improvement is the incorporation of more background knowledge for each factor. This additional context may help to mitigate the limitations of the current model and enhance its overall accuracy and effectiveness.

To the best of the authors' knowledge, we are the first one to present an approach to combine the method of Explainable AI (XAI) using SHAP with a conversational AI model, ChatGPT, to explain the predictions made by a nonlinear black box machine learning model. In recognition of the importance of this issue, MechaniX Limited is preparing a working paper that explores the potential for further advancements in prompt engineering for the ChatGPT model. The evolution of technology is poised to revolutionize the financial engineering landscape in profound ways. The imminent release of GPT-4, coupled with the capability to harness the power of natural language models and seamlessly integrate real-time updates from the internet emergence of a revolutionary new discipline - one that blends traditional quantitative analytics with human-interpretable models.

The result of our research has substantial significance for non-credentialed investors who possess limited financial literacy. In contrast to mutual funds, hedge funds are largely unregulated as unregistered private investment pools. [43]They are governed by their partnership agreement and make use of exemptions within the legal framework to exempt themselves from several regulations that apply to registered investment vehicles.[44] These exemptions include regulations pertaining to the composition of their portfolios and the nature of their investment strategies.[45] Given hedge funds are subject to significantly weaker investor protection regulations, it is reasonable to implement measures aimed at limiting share ownership to "accredited investors".[46]

[43] Sharma, Sunil. 1998. "Regulation of Hedge Funds." In Hedge Funds and Financial Market Dynamics, 166.
[44] Ibid.
[45] Ibid.
[46] According to the Securities Act of 1933, a domestic hedge fund that is structured as a 3(c)(1) fund may typically admit a maximum of 35 investors who do not meet the definition of "accredited investors". The remaining investors in the fund must generally be accredited investors, who are defined as individual investors with a net worth exceeding $1 million, excluding the value of their primary residence. Additionally, an individual earning a yearly income of over $200,000 or a combined income of $300,000 with a spouse, is typically considered to be an accredited investor.

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However, the decision is based on an implied assumption that "accredited investors" are sophisticated enough to "fend for themselves" in making investment decisions in highly complex unregistered financial securities, so that the information asymmetry between hedge funds and investors can be disregarded. [47]

IV. Conclusion

The recent advances in technology in Explainable AI (XAI) and language models such as ChatGPT have the potential to revolutionize the way in which financial knowledge is communicated to non-financial professionals. In our study, we tested the capabilities of ChatGPT in interpreting XAI models and its ability to explain complex financial concepts in a non-technical manner. The results show that ChatGPT has great potential as a tool for explaining complex financial concepts to a wide range of target audiences. The ability of ChatGPT to effectively transform financial knowledge into language that can be easily understood by non-financial professionals is a key advantage.

Although the current GPT-3.5 is not perfect in explaining model predictions yet, it is expected that the release of GPT-4 and the ability to harness the power of natural language models will revolutionize the financial engineering landscape and blend traditional quantitative analytics with human-interpretable models. Ultimately, the use of ChatGPT has the potential to level the playing field and empower all individuals, regardless of their financial background, to make informed investment decisions.

[47] Finke, Michael, Tao Guo, and Sandra Huston. 2021. "The unsophisticated "sophisticated": Old age and the accredited investors definition." Financial Planning Review 4, no. 2: e1114.

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